

Therapy/Counselling app

⚙ Status	Not started
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▼ Tech Stack (enterprise)

Service / Technology	Role in MVP	Phase Introduced	Approx. Monthly Cost (INR)
AWS EC2 (Worker Nodes)	Runs Kubernetes workloads (microservices)	Phase 0	₹5,000–₹7,000
Amazon EKS	Kubernetes control plane and orchestration	Phase 0	₹6,000–₹7,000
Amazon RDS (PostgreSQL)	Primary relational database (users, therapists, bookings, wallet, payments)	Phase 0	₹2,000–₹4,000
Amazon S3	Stores KYC documents, profile images, reports, attachments	Phase 0	₹100–₹300
Amazon ECR	Stores Docker container images	Phase 0	₹100–₹300
Amazon VPC	Private networking and security isolation	Phase 0	₹0
CloudWatch	Logging, monitoring and alerts	Phase 0	₹300–₹800
IAM	Identity and access management	Phase 0	₹0
Route 53	Domain and DNS management	Phase 0	₹100–₹200
AWS Certificate Manager (ACM)	SSL/TLS certificate management	Phase 0	₹0
GitHub Actions	CI/CD pipeline	Phase 0	₹0–₹1,000
Email Service (Resend/SES)	Email verification, OTPs, notifications	Phase 1	₹200–₹800
SMS/OTP Provider	Mobile OTP verification	Phase 1	₹500–₹2,000
KYC Verification API	Therapist identity verification	Phase 1	₹500–₹2,000
Amazon ElastiCache (Redis)	Caching, sessions, rate limiting	Phase 2	₹1,500–₹3,000
Payment Gateway (e.g., Stripe/Razorpay)	Wallet top-ups and consultation payments	Phase 3	No fixed monthly fee — see note below
MongoDB	Chat messages and real-time communication data	Phase 4	₹1,000–₹3,000
Daily.co / 100ms	Audio and video consultation infrastructure	Phase 4	₹2,000–₹8,000
RunPod L4	Runs YOLOv11, MediaPipe Hands, PaddleOCR and	Phase 5	₹4,400–₹6,500

Service / Technology	Role in MVP	Phase Introduced	Approx. Monthly Cost (INR)
	Whisper		
Together AI (Llama 3.1 8B API)	Stage 2 reasoning engine for moderation	Phase 5	₹200–₹1,000
YOLOv11	Detects phone screens, QR codes, ID cards, notes, etc.	Phase 5	Included in RunPod
MediaPipe Hands	Hand landmark detection and digit gesture recognition	Phase 5	Included in RunPod
PaddleOCR	OCR on suspicious video frames	Phase 5	Included in RunPod
Whisper (Self-hosted)	Speech-to-text transcription for video moderation	Phase 5	Included in RunPod
 Payment Gateway Transaction Fees	Per-transaction fee on every wallet top-up/payment	Phase 3	~2% + GST per transaction (variable, usage-dependent — not included in totals below)

Estimated recurring monthly infrastructure cost (during Phase 5 MVP development):
₹24,000–₹47,000/month, excluding payment gateway transaction fees and developer salaries.

 **Note on Payment Gateway costs:** the “No fixed monthly fee” line is misleading at scale — transaction fees (~2%+GST via Razorpay/Stripe) are usage-based and can exceed the entire infrastructure budget once volume grows. Track this separately as a % of revenue, not as a line item that reads as “free.”

 **Note on GST Reverse Charge (RCM):** RunPod, Together AI, and Daily.co/100ms are foreign vendors. Payments to them may attract GST under the Reverse Charge Mechanism. Flag this with your CA/tax advisor before Phase 5.

▼ Alternatives for MVP

Current Stack	Approx. Monthly Cost (INR)	Alternative Stack	Approx. Monthly Cost (INR)	Approx. Savings (INR/Month)	Risk Note
Amazon EKS	₹6,000–₹7,000	Docker Compose on EC2	₹0*	₹6,000–₹7,000	—
EC2 Worker Nodes (2× t3.medium)	₹5,000–₹7,000	Single EC2 (t3.large)	₹2,800–₹3,800	₹2,200–₹3,200	—
Amazon RDS (PostgreSQL)	₹2,000–₹4,000	PostgreSQL on EC2	₹0*	₹2,000–₹4,000	—
Amazon ElastiCache (Redis)	₹1,500–₹3,000	Redis on EC2	₹0*	₹1,500–₹3,000	—
MongoDB Atlas	₹1,000–₹3,000	MongoDB on EC2	₹0*	₹1,000–₹3,000	—
Amazon ECR	₹100–₹300	GitHub Container Registry	₹0	₹100–₹300	—
Amazon S3	₹100–₹300	Cloudflare R2	₹100–₹250	₹0–₹200	—
CloudWatch	₹300–₹800	Grafana + Prometheus + Loki (on EC2)	₹0*	₹300–₹800	—
Route 53	₹100–₹200	Cloudflare DNS	₹0	₹100–₹200	—
GitHub Actions (Paid)	₹0–₹1,000	GitHub Actions Free Tier	₹0	Up to ₹1,000	—
Daily.co / 100ms	₹2,000–₹8,000	Self-hosted LiveKit	₹500–₹2,000	₹1,500–₹6,000	 You own uptime/reliability for live therapy sessions — a dropped call mid-session is higher-stakes here than in a typical app. Requires dedicated DevOps + TURN/bandwidth cost not shown.
RunPod L4	₹4,400–₹6,500	RunPod T4	₹3,200–₹4,800	₹1,200–₹1,700	—

Current Stack	Approx. Monthly Cost (INR)	Alternative Stack	Approx. Monthly Cost (INR)	Approx. Savings (INR/Month)	Risk Note
Together AI (Llama 3.1 8B API)	₹200–₹1,000	Self-hosted Llama 3.1 8B**	₹4,500–₹6,500	-ve ₹400 to - ₹5,500	—
Email Service (Resend/SES)	₹200–₹800	Resend Free Tier	₹0	₹200–₹800	—
KYC Verification API	₹500–₹2,000	Manual Verification (Admin)	₹0	₹500–₹2,000	 Directly trades off against Loophole #1 (Customer–Therapist Collusion) and #7 (Fake Credential Fraud). Acceptable as a temporary MVP measure only — automate by Phase 8 as already planned, and don't leave manual verification running past that point.

Notes

- These services run on the same EC2 instance, so there is no separate infrastructure charge beyond the EC2 cost.
- *Self-hosting Llama removes API charges but increases GPU utilization. For your MVP, keeping **Together AI** is the more practical and cost-effective choice unless API usage becomes very high.

▼ Phases

▼ MVP Phases

Phase 0 – Foundation (Weeks 1-3)

Requirements lock, tech stack setup, repo/CI-CD scaffolding, cloud infra base (VPC, K8s cluster), auth service.

ENTERPRISE STACK

Service	Purpose	Approx. Monthly Cost (INR)
Amazon EKS	Kubernetes cluster	₹6,000–₹7,000
EC2 Worker Nodes	Run Auth service & backend	₹5,000–₹7,000
Amazon RDS (PostgreSQL)	Auth & user database	₹2,000–₹4,000
Amazon ECR	Docker image registry	₹100–₹300
Amazon VPC	Networking	₹0
IAM	Access management	₹0
AWS Certificate Manager	SSL certificates	₹0
Route 53	DNS	₹100–₹200
GitHub Actions	CI/CD	₹0–₹1,000
Estimated Total		₹13,200–₹19,500

MVP STACK

Current Service	Alternative	Monthly Cost (INR)
Amazon EKS	Docker Compose on EC2	₹0
EC2 Worker Nodes	Single EC2 (t3.large)	₹2,800–₹3,800
Amazon RDS (PostgreSQL)	PostgreSQL on EC2	₹0*
Amazon ECR	GitHub Container Registry	₹0
Route 53	Cloudflare DNS	₹0
Amazon VPC	Keep AWS VPC	₹0
IAM	Keep IAM	₹0
AWS Certificate Manager	Keep ACM	₹0
GitHub Actions	GitHub Actions Free Tier	₹0
Estimated Total		₹2,800–₹3,800

Phase 1 – Core Identity & Profiles (Weeks 4-7)

User + therapist onboarding, KYC/verification flow, profile management, admin panel v1 (approve therapists).

ENTERPRISE STACK

Service	Purpose	Approx. Monthly Cost (INR)
Existing Phase 0 Infrastructure	Backend	Included

Service	Purpose	Approx. Monthly Cost (INR)
Amazon S3	KYC documents & profile photos	₹100–₹300
CloudWatch	Logs & monitoring	₹300–₹800
Email Service	Email verification	₹200–₹800
SMS/OTP Provider	Phone verification	₹500–₹2,000
KYC Verification API	Therapist verification	₹500–₹2,000
Additional Cost		₹1,600–₹5,900

MVP STACK

Current Service	Alternative	Monthly Cost (INR)
Amazon S3	Cloudflare R2	₹100–₹250
CloudWatch	Grafana + Loki + Prometheus (on EC2)	₹0*
Email Service	Resend Free Tier	₹0
SMS/OTP Provider	Keep existing provider	₹500–₹2,000
KYC Verification API	Manual Admin Verification	₹0
Additional Cost		₹600–₹2,250

Phase 2 – Discovery & Booking (Weeks 8–11)

Search/matching (basic filters, skip full ES ranking for MVP), scheduling, calendar/availability, booking service.

ENTERPRISE STACK

Service	Purpose	Approx. Monthly Cost (INR)
Existing Infrastructure	Backend	Included
Amazon ElastiCache (Redis)	Cache & sessions	₹1,500–₹3,000
Additional Cost		₹1,500–₹3,000

MVP STACK

Current Service	Alternative	Monthly Cost (INR)
Amazon ElastiCache (Redis)	Redis on EC2	₹0*
Additional Cost		₹0

Phase 3 – Wallet & Payments (Weeks 12–14)

Wallet service, payment gateway integration, top-up/debit flows, basic ledger.

Service	Purpose	Approx. Monthly Cost (INR)
Existing Infrastructure	Backend	Included

Service	Purpose	Approx. Monthly Cost (INR)
Payment Gateway	Wallet & payments	No fixed monthly fee — NEW see transaction fee note above (~2%+GST)
Additional Cost		₹0 fixed

Phase 4 – Consultation Delivery (Weeks 15-19)

Chat (Socket.IO + Mongo), audio/video via Daily.co/100ms integration, session lifecycle management.

ENTERPRISE STACK

Service	Purpose	Approx. Monthly Cost (INR)
Existing Infrastructure	Backend	Included
MongoDB	Chat storage	₹1,000–₹3,000
Daily.co / 100ms	Audio & video sessions	₹2,000–₹8,000
Additional Cost		₹3,000–₹11,000

MVP STACK

Current Service	Alternative	Monthly Cost (INR)
MongoDB Atlas	MongoDB on EC2	₹0*
Daily.co / 100ms	Self-hosted LiveKit NEW see risk note above)	₹500–₹2,000
Additional Cost		₹500–₹2,000

Phase 5 – Trust & Safety (Weeks 20-22)

Rule-engine moderation (regex layer), reporting/flagging, basic fraud checks — defer heavy ML to post-MVP.

NEW **Add explicit deliverable:** Crisis-detection-to-human-handoff flow. This phase currently covers fraud/moderation only — it must also define what happens the moment self-harm language is detected (transcription flag → alert routing → human response SLA → helpline handoff).

ENTERPRISE STACK

Service	Purpose	Approx. Monthly Cost (INR)
Existing Infrastructure	Backend	Included
RunPod L4	YOLOv11, PaddleOCR, MediaPipe, Whisper	₹4,400–₹6,500
Together AI (Llama 3.1 8B API)	AI reasoning	₹200–₹1,000
Additional Cost		₹4,600–₹7,500

MVP STACK

Current Service	Alternative	Monthly Cost (INR)
RunPod L4	RunPod T4	₹3,200–₹4,800
Together AI (Llama 3.1 8B API)	Keep Together AI (recommended)	₹200–₹1,000
Additional Cost		₹3,400–₹5,800

Model	GPU Hours	Cost (INR)
YOLOv11 (training + inference testing)	40	₹1,466
MediaPipe Hands	20	₹733
PaddleOCR	20	₹733
Whisper (integration & evaluation)	40	₹1,466
Llama 3.1 8B (API)	0 GPU	₹0
Total	120 hrs	₹4,398

Running on RunPod - NVIDIA L4

Phase 6 – Hardening & Launch Prep (Weeks 23-26)

Load testing, observability (logging/monitoring wired up), security audit, beta rollout, bug fixes.

NEW **Add explicit checklist item:** Crisis SOP live dry-run with real on-call staff, completed before beta rollout begins.

Model	GPU Hours	Cost (INR)
YOLOv11 (benchmarking)	20	₹733
MediaPipe Hands	10	₹367
PaddleOCR	10	₹367
Whisper (load testing)	20	₹733
Llama 3.1 8B (API)	0 GPU	₹0
Total	60 hrs	₹2,200

▼ Post-MVP Phases

Phase 7 – Scale-Ready Data Layer (Months 7-8)

Introduce Elasticsearch for real search/faceted filters, Kafka event bus replacing simple queues, TimescaleDB for analytics/billing events.

Current (MVP)	Action	Replace With	Approx. Monthly Cost (INR)
Docker Compose on EC2	✗ Replace	Amazon EKS	₹6,000–₹7,000
Single EC2	✗ Replace	2–3 EC2 Worker Nodes	₹8,000–₹12,000
PostgreSQL on EC2	✗ Replace	Amazon RDS PostgreSQL	₹4,000–₹8,000
Redis on EC2	✗ Replace	Amazon ElastiCache Redis	₹2,000–₹5,000
GitHub Container Registry	✓ Keep	—	₹0
Cloudflare DNS	✓ Keep	—	₹0
Cloudflare R2	✓ Keep	—	₹100–₹500
Grafana + Prometheus + Loki	✓ Keep	—	₹0*

Phase 8 – Matching & ML (Months 8-10)

Go matching engine, collaborative-filtering recommendation model (FastAPI), fine-tuned PII detection model, Isolation Forest/XGBoost fraud detection.

Current (MVP)	Action	Replace With	Approx. Monthly Cost (INR)
Manual KYC	✗ Replace	Automated KYC Provider	₹2,000–₹10,000 (usage dependent)
RunPod L4	✓ Keep	—	₹4,400–₹6,500
Together AI (Llama API)	✓ Keep	—	₹500–₹2,000
LiveKit / Daily.co / 100ms	✓ Keep	—	Existing

NEW Confirms the Phase 1/MVP-stack decision to use manual KYC is explicitly temporary — this is the point where automated verification must land, closing the Loophole #7 exposure window.

Phase 9 – Advanced Moderation & AI Ops (Months 9-11)

Full rule-engine + LLM moderation pipeline, OCR-based image moderation (Texttract), automated escalation workflows.

Current	Action	Replace With	Approx. Monthly Cost (INR)
RunPod L4	✗ Replace	AWS EC2 G5 (A10G)	₹45,000–₹70,000

Current	Action	Replace With	Approx. Monthly Cost (INR)
Together AI (Llama API)	⚠️ Evaluate	Self-hosted Llama 3.1 8B on A10G (if API cost justifies)	Included in GPU cost
PaddleOCR	✅ Keep	—	Included
YOLOv11	✅ Keep	—	Included
MediaPipe Hands	✅ Keep	—	Included
Whisper	✅ Keep	—	Included

Phase 10 – Reliability & Multi-Region (Months 10-12)

Active-passive DR (Mumbai/Singapore), streaming replication, cross-region S3, Istio service mesh, full OTel/Jaeger tracing.

Current	Action	Replace With	Approx. Monthly Cost (INR)
MongoDB on EC2	❌ Replace	MongoDB Atlas	₹6,000–₹15,000
Single-region AWS	❌ Replace	Mumbai + Singapore Regions	+₹15,000–₹30,000
S3 (single region)	❌ Upgrade	Cross-Region Replication	₹2,000–₹5,000
LiveKit / Daily.co / 100ms	✅ Keep	—	Existing

Phase 11 – Platform Hardening (Months 11-13)

Auto-scaling tuning for 10K concurrent sessions, chaos testing, security pen-test, compliance audit (data privacy, HIPAA-equivalent).

Current	Action	Replace With	Approx. Monthly Cost (INR)
Existing Monitoring	❌ Upgrade	Managed Grafana / AMP / AMG (optional)	₹2,000–₹5,000
Existing Security	❌ Add	External Penetration Testing	₹50,000–₹2,00,000 (one-time)
Existing Infrastructure	✅ Keep	—	Existing

NEW Add explicit line: DPDP Act compliance audit (India-specific) — separate from generic “data privacy” language, since DPDP has its own notice, consent, and breach-reporting requirements.

Phase 12 – Growth Features (Months 12-15)

Push notifications at scale (FCM), advanced admin analytics dashboards, therapist performance insights, referral/loyalty in wallet, group sessions (stretch).

Current	Action	Replace With	Approx. Monthly Cost (INR)
FCM	✓ Keep	—	₹0
Analytics Dashboard	✓ Keep	—	Existing
Kafka	✓ Keep	—	Existing
Elasticsearch	✓ Keep	—	Existing

Phase 13 – Final Product Launch (Month 15-16)

Full regression, load test at target scale, marketing rollout, monitoring/alerting SLOs finalized, GA release.

Current	Action	Replace With	Approx. Monthly Cost (INR)
All Production Infrastructure	✓ Keep	—	Existing
Monitoring & Alerting	✓ Keep	—	Existing
AI Stack	✓ Keep	—	Existing
Multi-region Deployment	✓ Keep	—	Existing

▼ Revenue Model

This document has extensive cost detail but no revenue-side numbers, so per-session cost figures below can't be judged as healthy or unsustainable without this. Fill in before finalizing:

Item	Notes
Commission % (platform take rate)	e.g., 15–25% of session price is typical for service marketplaces
Therapist payout split	Amount/percentage paid out per session
Pricing model	Pay-per-session vs. subscription vs. wallet credits
Average session price (target)	Needed to compare against ₹9–26 infra cost/session below
Wallet float handling	Interest/liability treatment of pre-paid wallet balances
Payment gateway fee absorption	Who bears the ~2%+GST — platform, therapist, or customer

▼ Session Costing

▼ Per Session Cost

Assumptions

- 1 therapist + 1 user
- 60-minute video call
- Chat enabled
- AI moderation enabled (frame sampling every 2–3s)
- Whisper transcription
- Llama 3.1 8B reasoning only on flagged events
- Session metadata stored
- Typical AWS pricing and moderate production usage

Component	Cost / 60 min (INR)
Daily.co / 100ms Video Infrastructure	₹8.00–₹18.00
Backend APIs (EKS/EC2)	₹0.50–₹1.00
PostgreSQL (RDS)	₹0.10–₹0.30
Redis	₹0.05–₹0.15
MongoDB (Chat)	₹0.20–₹0.50
S3 Storage (attachments/logs)	₹0.05–₹0.10
AI GPU (YOLO + MediaPipe + PaddleOCR + Whisper)	₹1.50–₹3.50
Llama 3.1 8B API	₹0.05–₹0.30
Monitoring & Logging	₹0.20–₹0.50
Network/Data Transfer	₹0.50–₹2.00
 Payment Gateway Fee (~2%+GST of session price)	Variable — depends on session price, not fixed INR

 **Important:** all totals below are **infrastructure cost only**. They exclude payment gateway fees, therapist payout, and support/ops cost. Don't read "₹9–26/session" as "total cost to deliver a session" — see Revenue Model section above.

Total Infrastructure Cost

Scenario	Cost per 60-minute Session (INR)
Best Case	₹11–₹13
Typical	₹14–₹20
Heavy AI usage / higher video costs	₹20–₹26

▼ Per Session Cost (10 concurrent sessions)

Assuming:

- 10 concurrent 60-minute video sessions
- Your post-MVP architecture
- AI moderation enabled
- GPU utilization shared across the 10 sessions
- No idle infrastructure cost allocation (only operational cost during the session)

Component	Total Cost for 10 Concurrent Sessions (INR/hr)	Cost per Session (INR)
Daily.co / 100ms	₹80–₹180	₹8.0–₹18.0
Backend (EKS/EC2)	₹8–₹15	₹0.8–₹1.5
PostgreSQL (RDS)	₹2–₹4	₹0.2–₹0.4
Redis	₹1–₹2	₹0.1–₹0.2
MongoDB	₹2–₹5	₹0.2–₹0.5
S3	₹0.5–₹1	₹0.05–₹0.10
AI GPU (YOLO + MediaPipe + PaddleOCR + Whisper)	₹15–₹25	₹1.5–₹2.5
Llama 3.1 8B API	₹1–₹3	₹0.1–₹0.3
Monitoring & Logging	₹2–₹4	₹0.2–₹0.4
Network / Data Transfer	₹5–₹10	₹0.5–₹1.0

Total

Metric	Cost
Total for 10 concurrent 60-minute sessions	₹116–₹249
Average cost per 60-minute session	₹11.6–₹24.9

▼ Per Session Cost (100 concurrent sessions)

Component	Total Cost/Hour (INR)	Cost per Session (INR)
Daily.co / 100ms	₹800–₹1,800	₹8.0–₹18.0
Backend (EKS/EC2)	₹40–₹80	₹0.40–₹0.80
PostgreSQL (RDS)	₹10–₹20	₹0.10–₹0.20
Redis	₹5–₹10	₹0.05–₹0.10
MongoDB	₹15–₹30	₹0.15–₹0.30
S3	₹3–₹8	₹0.03–₹0.08
AI GPU Cluster	₹120–₹180	₹1.20–₹1.80
Llama 3.1 8B API	₹10–₹25	₹0.10–₹0.25
Monitoring & Logging	₹15–₹30	₹0.15–₹0.30
Network	₹40–₹80	₹0.40–₹0.80
Total	₹1,058–₹2,263	₹10.58–₹22.63

▼ Per Session Cost (1,000 concurrent sessions)

Component	Total Cost/Hour (INR)	Cost per Session (INR)
Daily.co / 100ms	₹8,000–₹18,000	₹8.0–₹18.0
Backend (EKS/EC2)	₹180–₹350	₹0.18–₹0.35
PostgreSQL (RDS)	₹40–₹80	₹0.04–₹0.08
Redis	₹20–₹40	₹0.02–₹0.04
MongoDB	₹50–₹100	₹0.05–₹0.10
S3	₹15–₹30	₹0.02–₹0.03
AI GPU Cluster	₹900–₹1,300	₹0.90–₹1.30
Llama 3.1 8B API	₹80–₹180	₹0.08–₹0.18
Monitoring & Logging	₹80–₹150	₹0.08–₹0.15
Network	₹250–₹500	₹0.25–₹0.50
Total	₹9,615–₹20,730	₹9.62–₹20.73

▼ Per Session Cost (10,000 concurrent sessions)

Component	Total Cost/Hour (INR)	Cost per Session (INR)
Daily.co / 100ms	₹80,000–₹180,000	₹8.0–₹18.0
Backend (EKS/EC2)	₹1,200–₹2,000	₹0.12–₹0.20
PostgreSQL (RDS)	₹250–₹500	₹0.03–₹0.05
Redis	₹100–₹200	₹0.01–₹0.02
MongoDB	₹300–₹600	₹0.03–₹0.06
S3	₹80–₹150	₹0.01–₹0.02
AI GPU Cluster	₹6,000–₹9,000	₹0.60–₹0.90
Llama 3.1 8B API	₹500–₹1,000	₹0.05–₹0.10
Monitoring & Logging	₹500–₹800	₹0.05–₹0.08
Network	₹1,500–₹2,500	₹0.15–₹0.25
Total	₹90,430–₹196,750	₹9.04–₹19.68

NEW **Note on video infra cost flatness:** Daily.co/100ms per-session cost (₹8–18) is shown as identical from 1 to 10,000 concurrent sessions. Video is usage-billed per participant-minute, so unlike compute/DB costs it typically does **not** get cheaper with scale, and may get worse once recording storage, bandwidth overages, and regional relay (TURN) costs are factored in at real scale. Re-verify this row against actual current Daily.co/100ms pricing tiers before relying on it.